

I225E Statistical Signal Processing

7. Spectral Analysis II

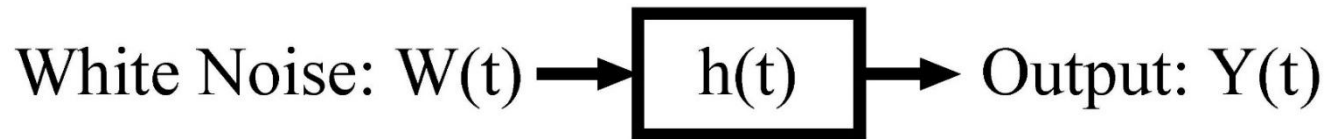
Candy Olivia MAWALIM

candylim@jaist.ac.jp

School of Information Science

White Noise and Linear System

- Consider white noise $W(t)$ as input to linear system, whose impulse response is given by $h(t)$.
- Find the output spectrum.



- Autocorrelation of white noise $W(t)$ are:

$$R_{WW}(\tau) = q\delta(\tau)$$

- **At $\tau = 0$:** $R_{WW}(0) = q\delta(0) \rightarrow \infty$ (conceptually, the value is very high). This implies that the signal is perfectly correlated with itself at the same time ($\tau = 0$).
- **At $\tau \neq 0$:** $R_{WW}(\tau) = q\delta(\tau) = 0$. This means that the white noise signal at any time t is completely uncorrelated with its value at any other time $t + \tau$ (where $\tau \neq 0$).

White Noise and Linear System

- Power-spectrum of white noise $W(t)$ are

$$S_{WW}(\omega) = q$$

- By the Wiener-Khinchin theorem:

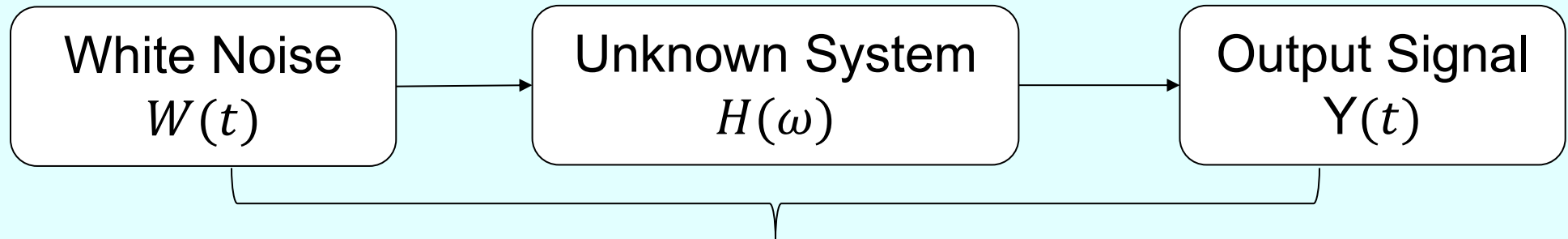
$$\begin{aligned} S_{WW}(\omega) &= F\{R_{WW}(\tau)\} = F\{q\delta(\tau)\} \\ &= q \int_{-\infty}^{\infty} \delta(\tau) e^{-j\omega\tau} d\tau = q e^{-j\omega(0)} = q \end{aligned}$$

- The power spectrum of white noise is a constant (q) across all frequencies ω (white noise has equal power at every frequency).
- Output Spectrum of an LTI System $S_{YY}(\omega)$ is

$$S_{YY}(\omega) = q|H(\omega)|^2$$

This implies that transfer function $H(\omega)$ can be obtained by injecting a white noise into linear system and then by computing the output spectrum. → **System Identification**

System Identification



Analysis and Identification

Identified System Characteristics: $(|H(\omega)|^2)$

■ Components:

■ Unknown System

- **Known Input** ← White noise is a particularly useful input for system identification because its power is evenly distributed across all frequencies.

■ Measured Output

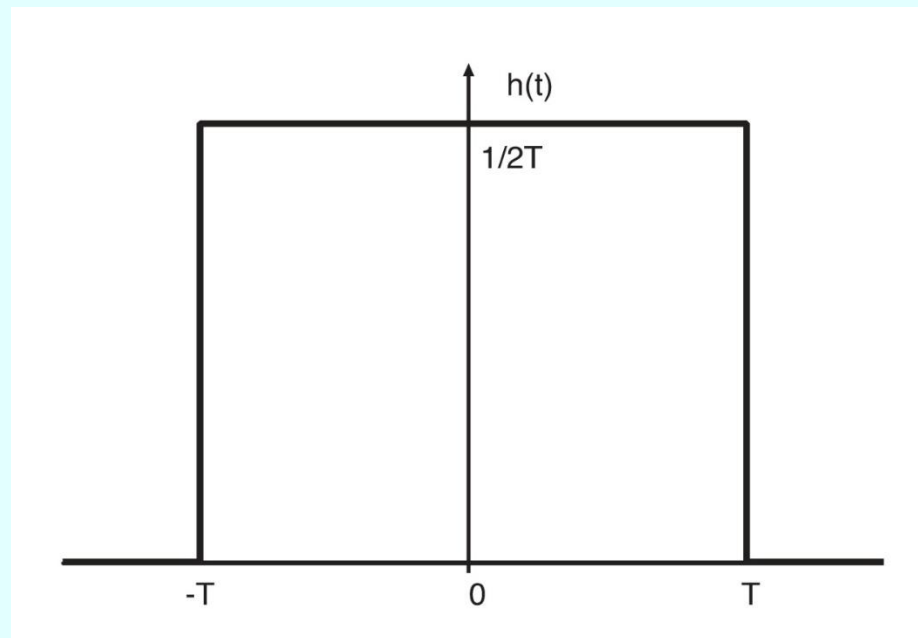
■ Inference of System Properties

Example 1: Smoothing Filter

- Linear system that takes moving average of input $X(t)$

$$Y(t) = \frac{1}{2T} \int_{t-T}^{t+T} X(\tau) d\tau$$

is called **smoothing filter**. Compute power-spectrum of output $Y(t)$.





Hint:

To compute the power spectrum of the output $Y(t)$, you need to:

1. Find the impulse response $h(t)$ of the moving average filter.
2. Compute the frequency response $H(\omega)$ by taking the Fourier Transform of $h(t)$.
3. Use the relationship $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$, where $S_{XX}(\omega)$ is the power spectrum of the input $X(t)$.

Denoting impulse response of linear system by $h(t)$,

$$Y(t) = \int_{-\infty}^{\infty} h(t - \tau)X(\tau)d\tau = h(t) * X(t)$$

$$H(\omega) = \int_{-T}^T \frac{1}{2T} e^{-i\omega t} dt = \frac{\sin T\omega}{T\omega}.$$

Therefore,

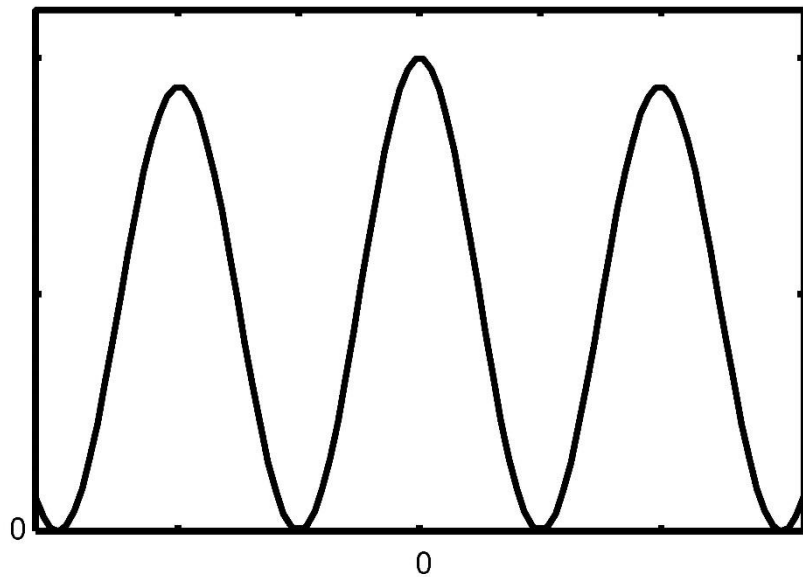
$$S_{YY}(\omega) = S_{XX}(\omega)|H(\omega)|^2 = S_{XX}(\omega) \frac{\sin^2 T\omega}{T^2 \omega^2}.$$

Smoothing filter functions as low-pass filter.

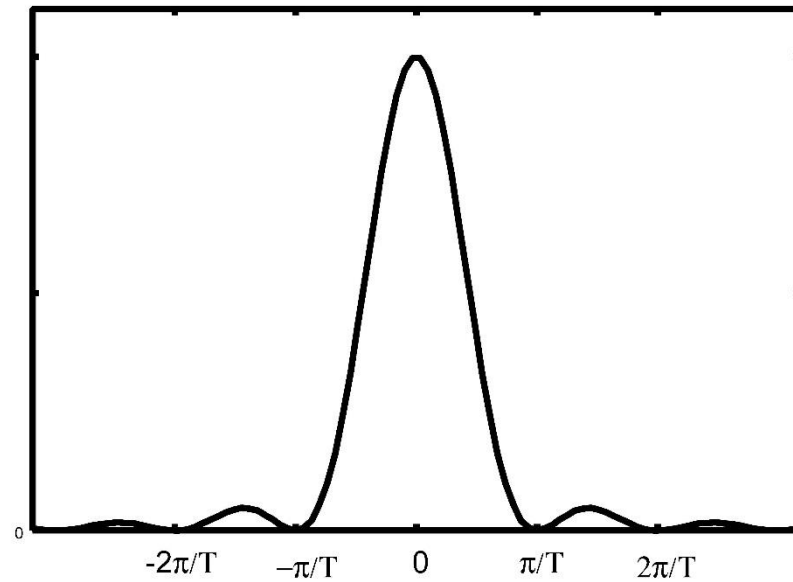
On the contrary, by subtracting the moving average from input $X(t)$ as

$$Z(t) = X(t) - \frac{1}{2T} \int_{t-T}^{t+T} X(\tau)d\tau$$

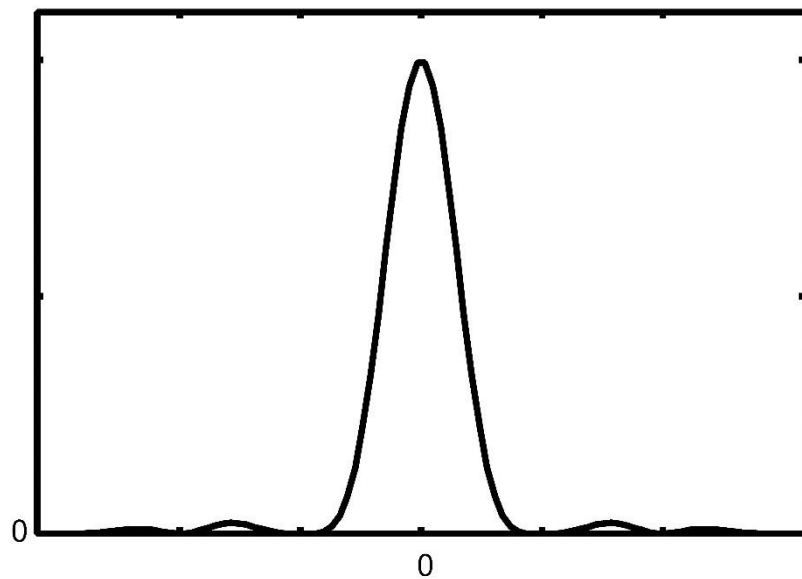
transfer function $H(\omega) = 1 - \frac{\sin T\omega}{T\omega}$ provides high-pass filter.



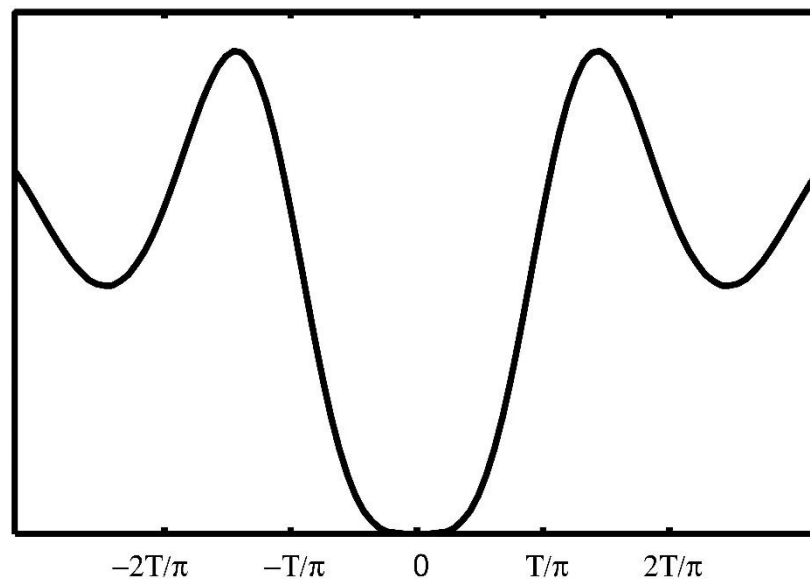
Input spectrum $S_{XX}(\omega)$



Transfer function $|H(\omega)|^2$



Output spectrum $S_{YY}(\omega)$



Example 2: Stochastic Resonance

- Input stationary process $X(t)$ into a linear system having the following transfer function:

$$H(\omega) = \frac{1}{\omega^2 - 2j\omega - 5}.$$

Suppose that input power is $E\{X^2(t)\} = 10$. Find input spectrum $S_{XX}(\omega)$ that maximizes power $E\{Y^2(t)\}$ of output $Y(t)$.

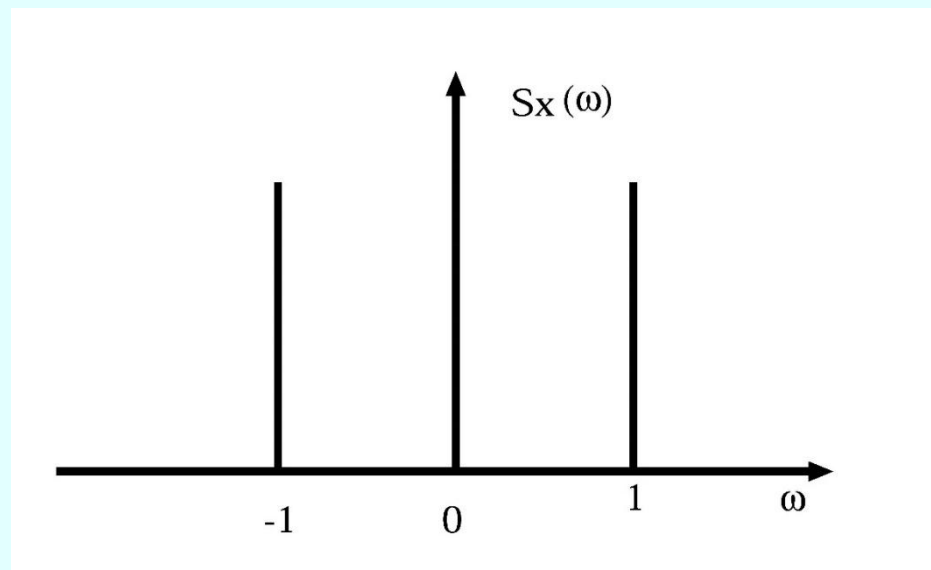
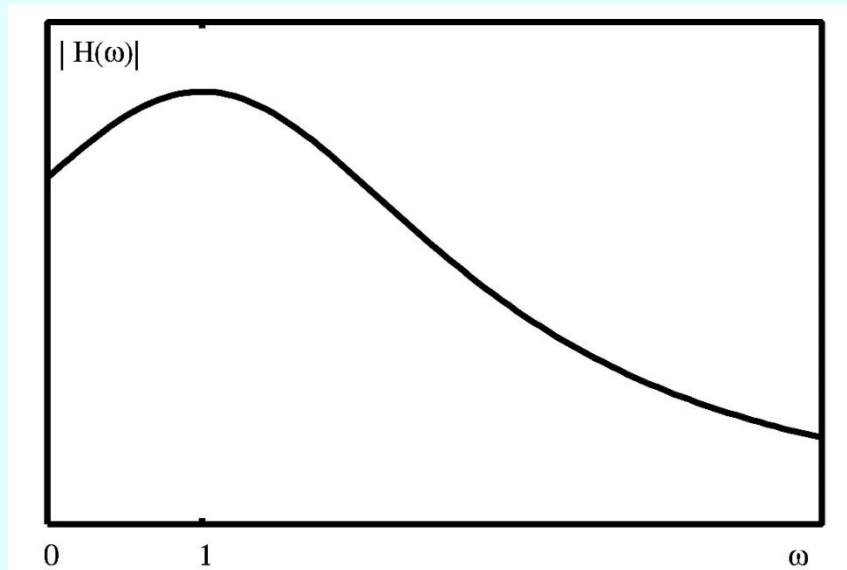
$$\begin{aligned} E\{Y^2(t)\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\omega) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) |H(\omega)|^2 d\omega \\ &\leq |H(\omega_n)|^2 \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \\ &= |H(\omega_n)|^2 E\{X^2(t)\} \end{aligned}$$

Here, $|H(\omega_n)|$ stands for maximal value of $|H(\omega)|$.

Because $|H(\omega)|^2 = H(\omega) \cdot H^*(\omega) = \frac{1}{\omega^4 - 6\omega^2 + 25}$, to find the minimum, we take the derivative of the denominator and set it to zero which makes $\omega^2 = 3$. So, $|H(\omega_n)|^2 = \frac{1}{16}$.

Hence,

$$E\{Y^2(t)\} \leq \frac{10}{16}. \text{ Equality holds for } R_{XX}(\tau) = 10 \cos \tau.$$



Example 3: Differential Equation

Consider a linear system with the following first-order linear differential equation with a white noise input $W(t)$:

$$Y'(t) + aY(t) = W(t)$$

where $a > 0$ is a constant.

$W(t)$ has an autocorrelation function $R_{WW}(\tau) = q\delta(\tau)$, where q is the power spectral density level.

Find:

- (a) The transfer function $H(\omega)$ of the system.
- (b) The power spectrum $S_{YY}(\omega)$ of the output $Y(t)$.
- (c) The autocorrelation function $R_{YY}(\tau)$ of the output $Y(t)$.



(a) The transfer function $H(\omega)$ of the system.

Take the Fourier Transform of the differential equation:

$$j\omega Y(\omega) + aY(\omega) = W(\omega)$$

$$Y(\omega)(j\omega + a) = W(\omega)$$

The transfer function $H(\omega) = \frac{Y(\omega)}{W(\omega)}$ is:

$$H(\omega) = \frac{1}{j\omega + a}$$

(b) The power spectrum $S_{YY}(\omega)$ of the output $Y(t)$.

First, find the power spectrum of the input $W(t)$:

$$S_{WW}(\omega) = q$$

The power spectrum of the output $Y(t)$ is given by:

$$S_{YY}(\omega) = |H(\omega)|^2 S_{WW}(\omega)$$

We need to find $|H(\omega)|^2$:

$$|H(\omega)|^2 = H(\omega)H^*(\omega) = \frac{1}{j\omega + a} \cdot \frac{1}{-j\omega + a} = \frac{1}{a^2 + \omega^2}$$

Now, substitute $S_{WW}(\omega) = q$:

$$S_{YY}(\omega) = \frac{1}{a^2 + \omega^2} \cdot q = \frac{q}{a^2 + \omega^2}$$

(c) The autocorrelation function $R_{YY}(\tau)$ of the output $Y(t)$.

The autocorrelation function $R_{YY}(\tau)$ is the inverse Fourier Transform of the power spectrum $S_{YY}(\omega)$:

$$R_{YY}(\tau) = F^{-1}\{S_{YY}(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{q}{a^2 + \omega^2} e^{j\omega\tau} d\omega$$

We know that the inverse Fourier Transform of $\frac{1}{a^2 + \omega^2}$ is $\frac{\pi}{a} e^{-a|\tau|}$.

Therefore:

$$R_{YY}(\tau) = \frac{q}{2\pi} \cdot \frac{\pi}{a} e^{-a|\tau|} = \frac{q}{2a} e^{-a|\tau|}$$

Exercise

For a system of differential equation with white noise input, find auto-correlation and power-spectrum of the system.

$$Y''(t) + 5Y'(t) + 6Y(t) = X(t),$$

$$R_{XX}(\tau) = 60\delta(\tau).$$

Solution: By substituting $t = t_2$, multiplying by $X(t_1)$ from left hand side, and taking expectation,

$$R_{XY''}(t_1, t_2) + 5R_{XY'}(t_1, t_2) + 6R_{XY}(t_1, t_2) = 60\delta(\tau)$$

By denoting $\tau = t_1 - t_2$ and taking Fourier transform

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau,$$

$$\begin{aligned}
S_{XY'}(\omega) &= \int_{-\infty}^{\infty} R_{XY'}(\tau) e^{-i\omega\tau} d\tau \\
&= \int_{-\infty}^{\infty} (-1) \frac{dR_{XY}(\tau)}{d\tau} e^{-i\omega\tau} d\tau \\
&= (-i\omega) \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau \\
&= (-i\omega) S_{XY}(\omega) \\
S_{XY''}(\omega) &= (-i\omega) S_{XY'}(\omega) = (-i\omega)^2 S_{XY}(\omega)
\end{aligned}$$

Therefore,

$$\begin{aligned}
(-i\omega)^2 S_{XY}(\omega) + (-i\omega) 5 S_{XY}(\omega) + 6 S_{XY}(\omega) &= 60 \\
\rightarrow S_{XY}(\omega) &= \frac{60}{-\omega^2 - 5i\omega + 6}
\end{aligned}$$

By taking the Fourier transform of

$$R_{Y''Y}(\tau) + 5R_{Y'Y}(\tau) + 6R_{YY}(\tau) = R_{XY}(\tau),$$

$$(i\omega)^2 S_{YY}(\omega) + 5(j\omega) S_{YY}(\omega) + 6S_{YY}(\omega) = \frac{60}{-\omega^2 - 5j\omega + 6}.$$

Hence,

$$S_{YY}(\omega) = \frac{60}{(-\omega^2 + 5i\omega + 6)(-\omega^2 - 5i\omega + 6)}$$

$$= 3 \frac{2 \times 2}{2^2 + \omega^2} - 2 \frac{2 \times 3}{3^2 + \omega^2}$$

$$R_{YY}(\tau) = 3e^{-2|\tau|} - 2e^{-3|\tau|}$$

Application of Differential Equation

- $X(t)$: Location of particle,
- m : Mass
- f : Coefficient of friction
- $cX(t)$: External force
(c is constant and constrained motion if $c \neq 0$)
- $F(t)$: Collision force
- T : Temperature
- k : Boltzmann constant

$$mX''(t) + fX'(t) + cX(t) = F(t)$$

Suppose $F(t)$ has mean $E\{F(t)\} = 0$ and power-spectrum $S_F(\omega) = 2kTf$.

Solution:

The system function is

$$\begin{aligned}\frac{1}{|H(\omega)|^2} &= (ms^2 + fs + c)(ms^2 - fs + c) \big|_{s=i\omega} \\ &= (c - m\omega^2)^2 + f^2\omega^2.\end{aligned}$$

Therefore,

$$\begin{aligned}S_X(\omega) &= S_F(\omega)|H(\omega)|^2 \\ &= \frac{2kTf}{(c - m\omega^2)^2 + f^2\omega^2}.\end{aligned}$$

If equation $ms^2 + fs + c = 0$ has complex conjugate

$$s_{1,2} = -\alpha \pm i\beta,$$

$$\alpha = \frac{f}{2m}, \quad \alpha^2 + \beta^2 = \frac{c}{m},$$

autocorrelation of $X(t)$ is

$$R_X(\tau) = \frac{kT}{c} e^{-\alpha|\tau|} (\cos \beta\tau + \frac{\beta}{\alpha} \sin \beta|\tau|).$$

Since $X(t)$ is a normal process with mean 0 and variance $R_X(0) = \frac{kT}{c}$ for fixed t , its density is

$$f_X(x) = \sqrt{\frac{c}{2\pi kT}} e^{-\frac{cx^2}{2kT}}$$

In case of Free Motion ($c = 0$):

Denoting $V(t) = X'(t)$, $mV'(t) + fV(t) = F(t)$.

The system function is

$$\frac{1}{|H(\omega)|^2} = (ms + f)(ms - f) \big|_{s=j\omega} = m^2\omega^2 + f^2.$$

Therefore, $S_V(\omega) = S_F(\omega)|H(\omega)|^2 = \frac{2kTf}{m^2\omega^2 + f^2},$

$$R_V(\tau) = \frac{kT}{m} e^{-\frac{f|\tau|}{m}}.$$

Since $V(t)$ is a normal process with mean 0 and

variance $R_V(0) = \frac{kT}{m}$, its density is $f_V(v) = \sqrt{\frac{m}{2\pi kT}} e^{-\frac{mv^2}{2kT}}$

Causality

■ Definition:

A system is said to be causal if its output at any time t depends only on the input at the present time t and in the past (i.e., for times $\leq t$).

■ In simpler terms, a causal system's output cannot "predict" future inputs.

■ Real-world physical systems are generally causal because an effect cannot precede its cause.

Causality

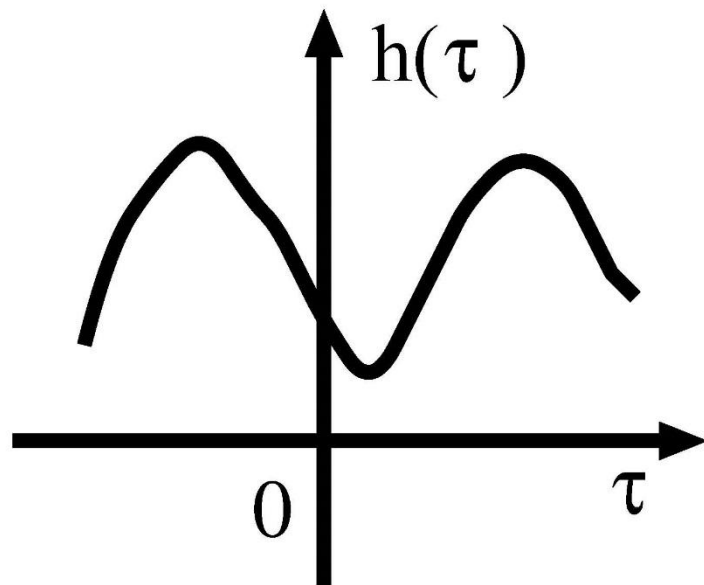
- The impulse response $h(\tau)$ of an LTI system completely characterizes the system.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

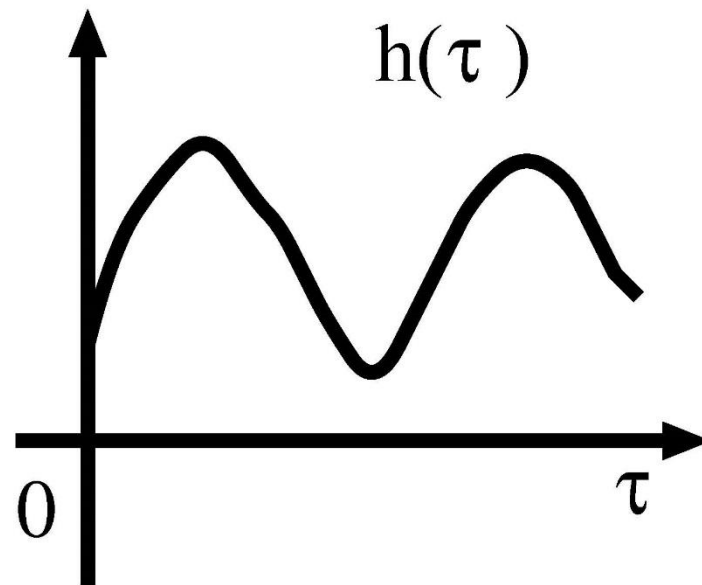
- For causality, $y(t)$ should only depend on $x(\tau)$ where $\tau \leq t$.
- This means that if $\tau > t$ (future input), the contribution to the integral should be zero. This happens if $h(t - \tau) = 0$ when $\tau > t$.
- Let $\alpha = t - \tau$. If $\tau > t$, then $\alpha < 0$. So, for causality, we need $h(\alpha) = 0$ for all $\alpha < 0$.

Causality

Non-causal



Causal



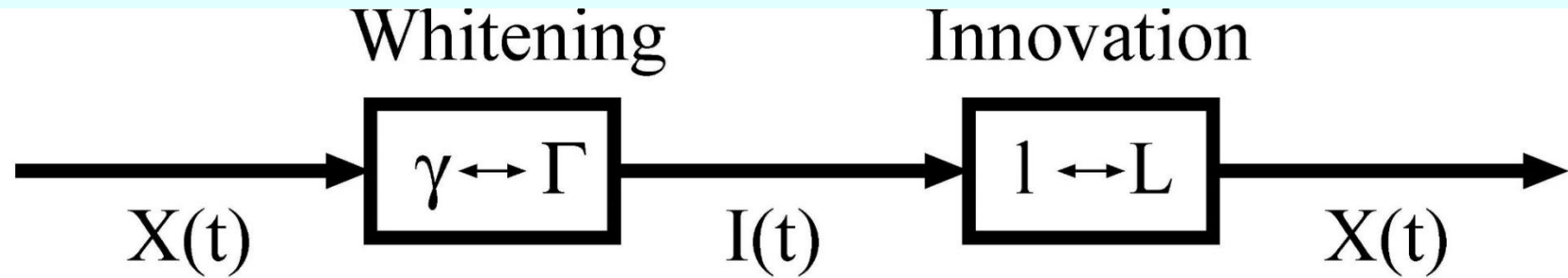
- Non-causal: For $\tau < 0$, $h(\tau) \neq 0$.
- Causal: For $\tau < 0$, $h(\tau) = 0$.

Exercise

Determine whether the following LTI systems, defined by their impulse responses $h(t)$, are causal or non-causal. Justify your answer for each system.

- a) $h_1(t) = e^{-2t}u(t)$, where $u(t)$ is the unit step function ($u(t) = 1$ for $t \geq 0$, and $u(t) = 0$ for $t < 0$).
- b) $h_2(t) = e^{3t}u(-t)$, where $u(-t) = 1$ for $t \leq 0$, and $u(-t) = 0$ for $t > 0$.
- c) $h_3(t) = \sin(t)$ for all t .

Factorization and Innovation



- Stationary process $X(t)$ (Input to Whitening)
- "Whitening" Block ($\gamma \leftrightarrow \Gamma$):
 - $\Gamma(s)$: Whitening filter
 - This block represents a filter or an operation that aims to transform the input $X(t)$ into **an innovations of $X(t)$** , denoted as $I(t)$, e.g., a white noise process.
- "Innovation" Block ($I \leftrightarrow L$): **$L(s)$: Innovation filter**
 - This block represents a shaping filter or an operation that takes the innovation process and transforms it back into $X(t)$.

Factorization and Innovation

For stationary process $X(t)$, consider system $\Gamma(s)$ that satisfies

$$I(t) = \int_0^\infty \gamma(\alpha) X(t - \alpha) d\alpha$$

$$R_{II}(\tau) = E\{I(t + \tau)I(t)\} = \delta(\tau)$$

Since $L(s) = 1/\Gamma(s)$ is stable and causal,

$$X(t) = \int_0^\infty l(\alpha) I(t - \alpha) d\alpha.$$

$X(t)$ is called **regular**.



Note Laplace transform:

$$\Gamma(s) = \int_0^{\infty} \gamma(\tau) e^{-s\tau} d\tau$$

$$L(s) = \int_0^{\infty} l(\tau) e^{-s\tau} d\tau$$

Necessary and sufficient condition of regular process

■ Denoting power-spectrum of $X(t)$ by $S_{XX}(s)$ ($s = i\omega$),

$$\begin{aligned} S_{XX}(i\omega) &= S_{II}(i\omega)|L(i\omega)|^2 \\ &= S_{II}(i\omega)L(i\omega)L^*(i\omega) \\ &= S_{II}(i\omega)L(i\omega)L(-i\omega). \end{aligned}$$

Since power-spectrum of $I(t)$ is $S_{II}(i\omega) = 1$, substitution of $s = i\omega$ yields $S_{XX}(s) = L(s)L(-s)$.

On the other hand, if power-spectrum of $X(t)$ can be written as the above product, power spectrum of $I(t)$ is

$$\begin{aligned} S_{II}(s) &= S_{XX}(s)\Gamma(s)\Gamma(-s) \\ &= L(s)L(-s)\Gamma(s)\Gamma(-s) = 1. \end{aligned}$$

Hence, necessary and sufficient condition for $X(t)$ to be regular is $S_{XX}(s)$ can be decomposed as

$$S_{XX}(s) = L(s)L(-s)$$

■ Paley-Wiener Condition

Necessary and sufficient condition for $X(t)$ to be regular is

$$\int_{-\infty}^{\infty} \frac{\ln S_{XX}(\omega)}{1 + \omega^2} d\omega < \infty$$

■ Rational Spectra

Any positive rational spectrum $S(\omega) = \frac{A(\omega)}{B(\omega)}$ (where $A(\omega)$ and $B(\omega)$ are polynomials of ω) satisfies the Paley-Wiener condition as

$$\int_{-\infty}^{\infty} \frac{\ln S_{XX}(\omega)}{1 + \omega^2} d\omega = \int_{-\infty}^{\infty} \frac{\ln A(\omega) - \ln B(\omega)}{1 + \omega^2} d\omega < \infty$$

Corresponding process $X(t)$ is regular.

■ In case $X(t)$ is real and has rational spectrum

Since $S_{XX}(-\omega) = S_{XX}(\omega)$, $S_{XX}(\omega) = A(\omega^2)/B(\omega^2)$.

Substitution of $s = j\omega$ yields

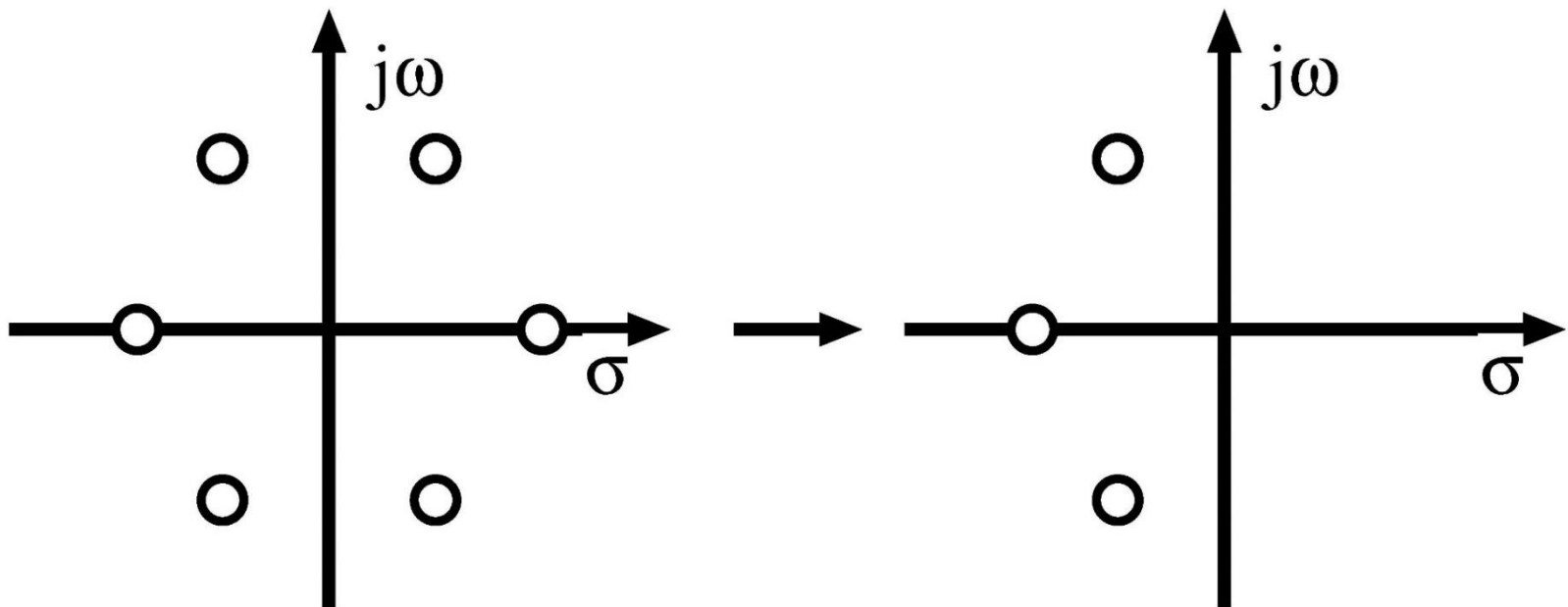
$$S_{XX}(s) = A(-s^2)/B(-s^2).$$

Since $A(-s^2)$ and $B(-s^2)$ have real coefficients, their roots s_i are real or complex conjugate. If s_i is pole or zero, $-s_i$ is also pole or zero.

All roots (pole or zero) can be separated into two groups: negative real part ($\Re s_i < 0$) and positive real part ($\Re s_i > 0$). $L(s)$ can be constructed using roots with negative real part as

$$S_{XX}(s) = \frac{N(s)N(-s)}{D(s)D(-s)}, \quad L(s) = \frac{N(s)}{D(s)}.$$

If $L(s)$ is analytic in space of negative real part ($\Re s_i < 0$), we say “ $L(s)$ has **minimum-phase property**.”



Matched Filter

- A matched filter is a linear time-invariant (LTI) filter designed **to maximize the signal-to-noise ratio (SNR) at its output at a specific time instant when the input to the filter is the signal $f(t)$ corrupted by additive noise $n(t)$.**
- Why is it called a "matched" filter?
The filter's impulse response is "matched" to the shape of the signal you are trying to detect (in a time-reversed and possibly conjugated sense), taking into account the characteristics of the noise.

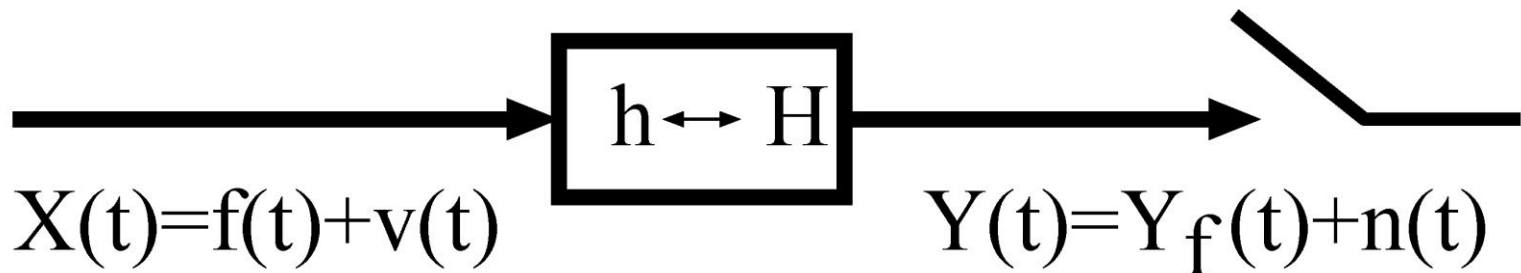
Matched Filter

Consider deterministic signal $f(t)$ with additive noise $v(t)$:

$$X(t) = f(t) + v(t)$$

We detect whether signal $f(t)$ is included in observation data $X(t)$. The function $f(t)$ is known; $v(t)$ is stationary noise and its power-spectrum $S_{vv}(\Omega)$ is known.

(Example: Detection of reflection wave from radar)





Consider $X(t)$ as input to system $H(\omega)$. Denoting the output by $Y(t)$,

$$Y(t) = Y_f(t) + \mathbf{n}(t),$$

$$Y_f(t) = f(t) * h(t),$$

$$\mathbf{n}(t) = \mathbf{v}(t) * h(t).$$

Signal-to-noise ratio (SNR) r_0 is given by

$$\begin{aligned} r_0 &= \frac{|Y_f(t)|^2}{E\{n^2(t)\}} = \frac{|Y_f(t)|^2}{\frac{1}{2\pi} \int_{-\infty}^{\infty} S_{nn}(\omega) d\omega} \\ &= \frac{|Y_f(t)|^2}{\frac{1}{2\pi} \int_{-\infty}^{\infty} S_{vv}(\omega) |H(\omega)|^2 d\omega} \end{aligned}$$

Design filter $H(\omega)$ in such a way that SNR is maximized.

Case of white noise

■ White noise $v(t)$

Because of white noise, $S_{vv}(\omega) = S_0$.

By inverse Fourier transform, $y_f(t)$ can be written as

$$y_f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) H(\omega) e^{i\omega t} d\omega.$$

From Cauchy-Schwarz inequality,

$$r_0 = \frac{\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) H(\omega) e^{i\omega t} d\omega \right|^2}{\frac{S_0}{2\pi} \int_{-\infty}^{\infty} |H(\omega)|^2 d\omega} \leq \frac{\int_{-\infty}^{\infty} |F(\omega)|^2 d\omega \int_{-\infty}^{\infty} |H(\omega)|^2 d\omega}{2\pi S_0 \int_{-\infty}^{\infty} |H(\omega)|^2 d\omega}$$



The equality holds in the following case:

$$H(\omega) = F^*(\omega)e^{-i\omega t},$$

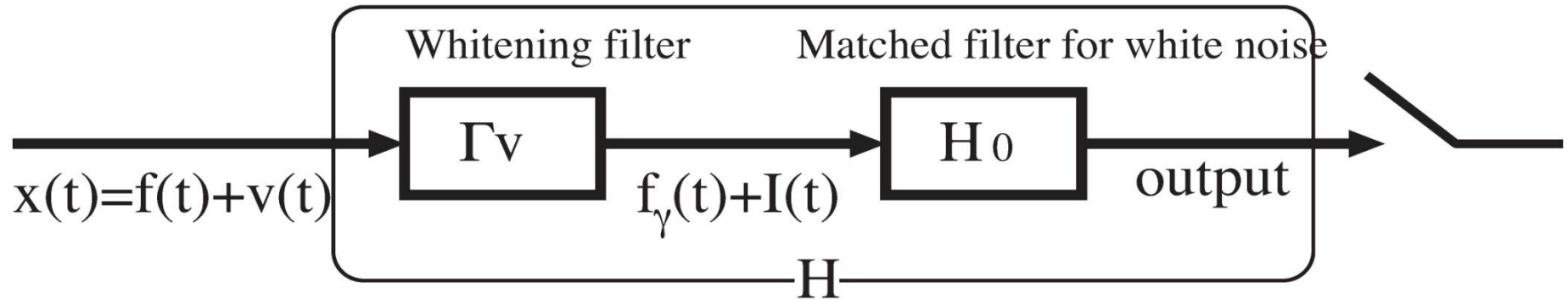
$$h(\tau) = f(t - \tau).$$

$H(\omega)$ that maximizes cross-correlation between $x(t) = f(t) + v(t)$ and $kf(t)$ is called **Matched Filter**.

Remark:

$h(\tau) = f(t - \tau)$ is not necessarily causal. In that case, optimal causal filter is given by $h(\tau) = f(t - \tau)U(\tau)$ (where $U(\tau) = 1$ ($\tau \geq 0$); 0 ($\tau < 0$)).

Case of colored noise



■ Colored noise: $v(t)$

Consider a whitening filter $\Gamma_v(j\omega)$ for $v(t)$. If we input $X(t) = f(t) + v(t)$ into $\Gamma_v(j\omega)$, the output is given by

$$\begin{aligned} Y(t) &= (f(t) + v(t)) * \gamma_v(t) \\ &= f(t) * \gamma_v(t) + v(t) * \gamma_v(t) = f_\gamma(t) + I(t) \end{aligned}$$

The output $Y(t)$ is a transformed deterministic signal $f_\gamma(t) = f(t) * \gamma_v(t)$ with additive white noise $I(t)$.

Using the results obtained with white noise, the optimal filter is given by

$$h_0(\tau) = f_\gamma(t - \tau),$$

$$H_0(\omega) = F_\gamma^*(\omega)e^{-i\omega t} = F^*(\omega)\Gamma_v^*(j\omega)e^{-i\omega t}.$$

Finally, the total filter is obtained as

$$\begin{aligned} H(\omega) &= H_0(\omega)\Gamma_v(i\omega) = F^*(\omega)\Gamma_v^*(i\omega)\Gamma_v(i\omega)e^{-i\omega t} \\ &= F^*(\omega)\left|\Gamma_v(i\omega)\right|^2 e^{-i\omega t} \\ &= \frac{F^*(\omega)e^{-i\omega t}}{S_{vv}(\omega)}. \end{aligned}$$